

E-Commerce Product Entity-Resolution & Matching Platform

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

This project presents the design and implementation of an E-Commerce Product Entity-Resolution and Matching Platform in Java, developed as part of the Advanced Data Structures and Algorithms curriculum. The platform addresses the real-world problem of identifying, deduplicating, and matching product listings that refer to the same entity across multiple e-commerce datasets, despite variations in product names, descriptions, and attributes a challenge actively faced by platforms such as Amazon, Flipkart, and Walmart.

The system is built entirely without the use of `java.util.*` inside the core engine, with every algorithm and data structure hand-implemented from scratch. Product name and description matching is performed using KMP and Rabin-Karp rolling hash for fast pattern search, and Z-function based suffix arrays for substring similarity computation. Fuzzy matching between product titles is achieved through the Wagner-Fischer implementation of Levenshtein Edit Distance and Damerau-Levenshtein Distance, directly applying the Advanced DP module. Jaccard Similarity over tokenized product descriptions is computed using a hand-built inverted index with a suffix array backed LCP structure for candidate blocking.

A graph is constructed where each product listing is a node and weighted similarity scores form edges. Union-Find (Disjoint Set Union) clusters matched products into resolved entities, while a hand-built min-heap priority queue ranks candidate pairs by match confidence. The Miller-Rabin primality test is applied in the polynomial rolling hash module for double-hashing collision avoidance, incorporating the Randomised Algorithms module. The pipeline ingests raw CSV product data, performs tokenization and stop-word removal using custom trie and hash map structures, and outputs a deduplicated, entity-resolved product catalogue.

This project demonstrates practical application of String Algorithms, Advanced Dynamic Programming, Graph-based clustering, and Randomised Hashing to a meaningful industry problem, with full asymptotic complexity analysis of each component across the resolution pipeline.

Signature of the Guide